ASEAN
GEOAI FUSION 2026

TRAINER / SPEAKER PRESENTATION

Introduction to Google Earth Engine (GEE)

Daniel Marc dela Torre, PhD | Science and Data Lead, Spatial Informatics Group

15 July 2026 • Bootcamp 1 / Spotlight Series • ASEAN GeoAI Fusion 2026

About the Trainer



Daniel Marc dela Torre, PhD

Science and Data Lead, Spatial Informatics Group

Daniel dela Torre, PhD, specializes in remote sensing for environmental monitoring, with expertise in machine learning and time series analysis. He earned his PhD in Geography from the University of Auckland, focusing on climate change impacts on rice yields. Previously, he led the geospatial team in a startup working on the utilities sector.

Geospatial AI | Remote Sensing / Geoinformatics | Environmental and Utilities Monitoring

About Spatial Informatics Group (SIG)

SIG
Spatial Informatics Group

We Translate Data Into Knowledge



Spatial Informatics Group (SIG) is a scientific think tank and boundary organization.

We convert geospatial data into impactful products and services, empowering decision-makers to effect positive change on a global scale. Supported by our four pillars—international talent, strategic partnerships, vertical integration, and commitment to open science—our organizational structure promotes agile collaboration among research scientists in specialized working groups.

ABOUT US

SIG
Spatial Informatics Group

What does SIG work on?

Forest & Agriculture Management

- Inventory
- Planning
- Biomass
- Bioenergy and Forest Product Assessment
- Forest Economics
- Forest Sustainability Assessment & Certification
- Forest Inventory and Biometrics
- Wildlife Inventory
- Monitoring and Vulnerability Assessment

Natural Hazards

- Disaster Risk Reduction
- Risk and Hazard Assessment
- Emissions and Weather Modeling
- Wildfire Monitoring and Predictions
- Flood Monitoring and Predictions
- Drought Monitoring and Predictions

Environmental Mapping

- Regional Land Cover Monitoring Systems
- Land Use Planning
- Data Collection and Preprocessing
- Feature extraction.
- Near Real Time alerts
- Crop Mapping
- Remote Sensing Based Forest Inventories
- Geospatial Data Integration
- ESG monitoring

Forest Carbon Offsets

- Feasibility Analytics
- Offset Generation
- Monitoring Reporting Verification
- Methodology Development

Climate Change

- Analysis and Mapping
- Risk Assessment and Mitigation
- Adaptation Planning

Ecosystem Services

- Scoping
- Quantification
- Planning and Management
- Markets and Payments
- Indicators

Google Cloud Partner

Session Agenda

Current Applications of GeoAI in the Telecommunications and Associated Industries

01	**Overview of Google Earth Engine**	20 min
02	**Sign-up to GEE**	10 min
03	**GEE Code Editor and Basics of Javascript**	30 min
04	**Example Applications of GEE**	10 min
05	**Q&A**	20 min

WHAT IS EARTH ENGINE?

1. Remote Sensing Archive with petabytes of data in one location
2. A cloud-based geospatial processing platform for executing large-scale data analysis.

Photo courtesy of Google Earth Outreach

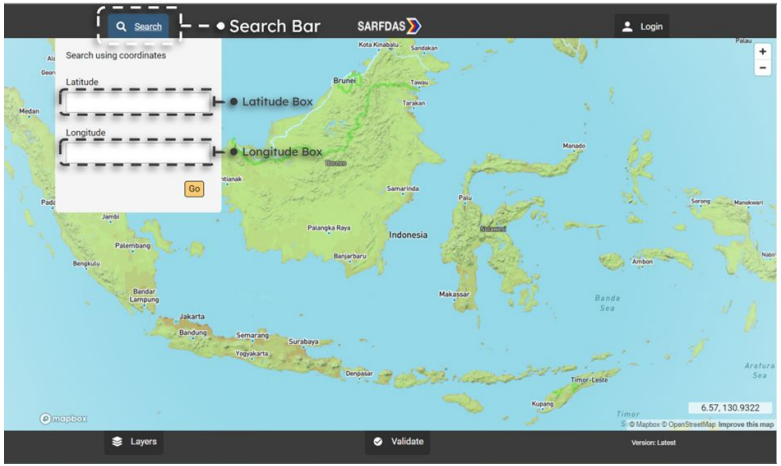


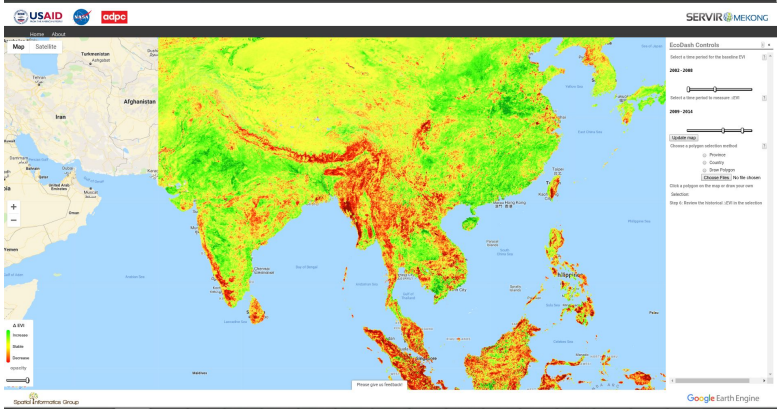
WHAT IS EARTH ENGINE?

Example Applications

Classifying land cover conditions and change detection:

- Mekong Regional Land Cover Monitoring System





WHAT IS EARTH ENGINE?

Why is it such a powerful image analysis software?

1. Public data catalog: vast amounts of publicly available data (you don’t need to store data)
2. Processing power (computation engine)
3. Interactive development platforms
4. Save and share work routines

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WHAT IS EARTH ENGINE?

Data Catalog

> hundreds public datasets
> thousands new images every day
> millions of images
> 100 petabytes of data

MODIS
250m daily

Landsat & Sentinel
10-30m, 14-day

Terrain &
Land Cover

Weather & Climate
NOAA NCEP, OMI, ...

... and many more, updating daily!

Source: Google Earth Engine Slides

EARTH ENGINE PUBLIC DATA CATALOG

• Imagery	• Climate & Weather
• Geophysical	• Demographic
• Vector Data	

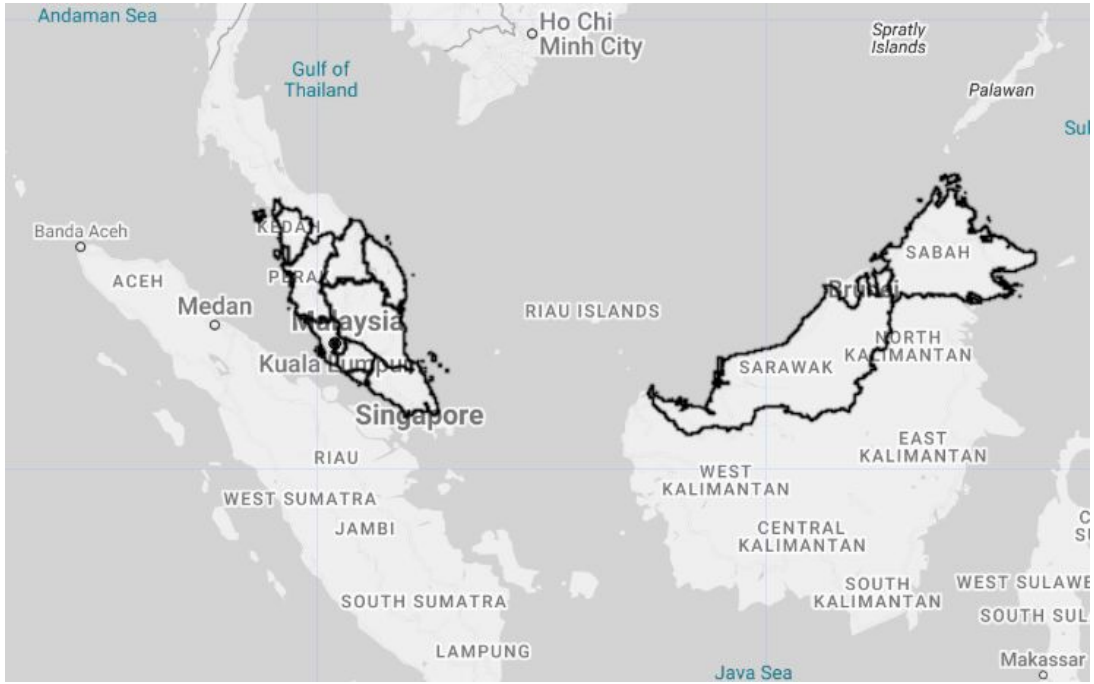


Image: Dave Thau

WHAT IS EARTH ENGINE?

Load your own data

- Shapefiles



https://code.earthengine.google.com/04ccae1d5a1cea9ede17981fde9f4844

WHAT IS EARTH ENGINE?

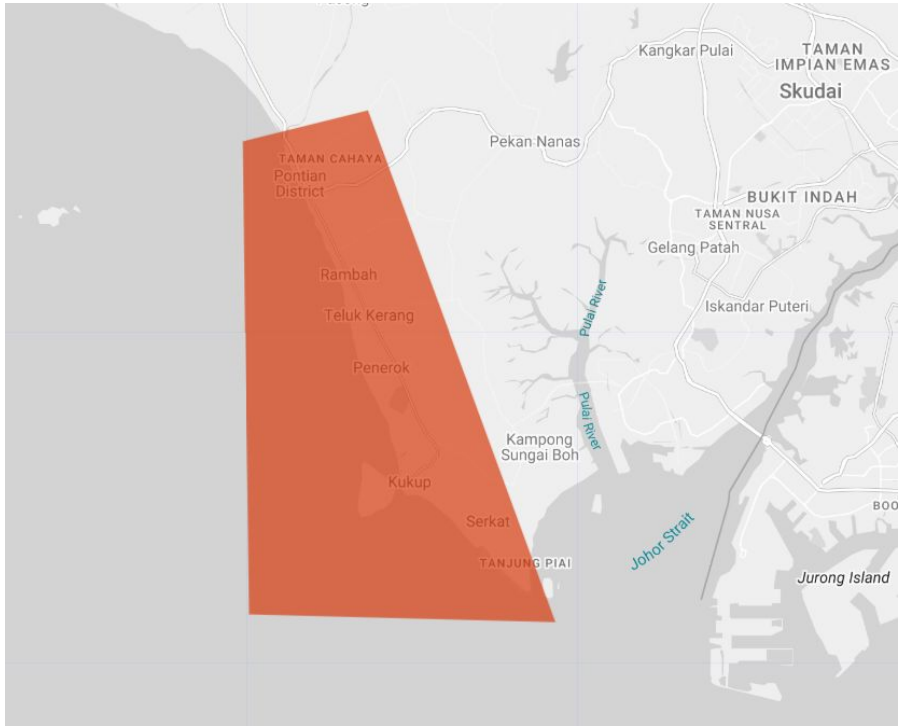
Make your own data in the map

Imports generated code

Below is the JavaScript code representing the current imports. To transfer them to another script, paste this code into the editor and click "Convert" in the suggestion tooltip.

```
var geometry = /* color: #d63000 */ee.Geometry.Polygon(
        [[[103.36092284157547, 1.2563111353348193],
          [103.52365782692704, 1.2521922493876252],
          [103.42409422829422, 1.5240238198485],
          [103.35748961403641, 1.507550094314364]]]);
```

CLOSE



WHAT IS EARTH ENGINE?

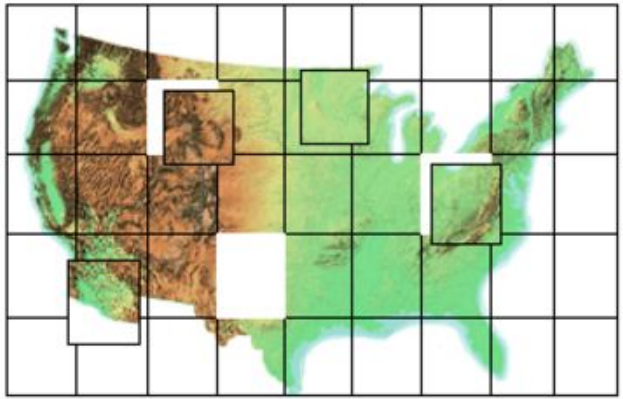
Why is it such a powerful image analysis software?

1. Public data catalog: vast amounts of publicly available data (you don’t need to store data)
2. Processing power (computation engine)
3. Interactive development platforms
4. Save and share work routines

COMPUTATION EARTH ENGINE

"The computation engine is a just in time distributed computation model, …"

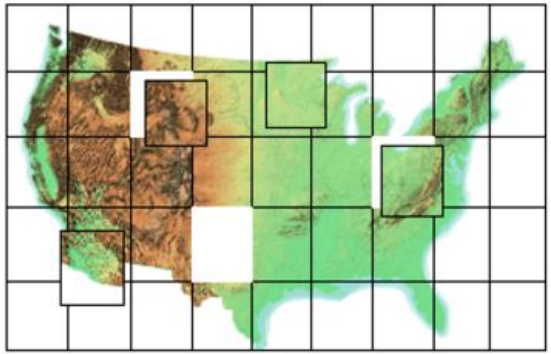
Divides data in independent grids



COMPUTATION EARTH ENGINE

"The computation engine is a just in time distributed computation model, a cloud-based processing infrastructure that automatically parallelized analyses on many CPUs across many computers in Google's data centers."

Divides data in independent grids

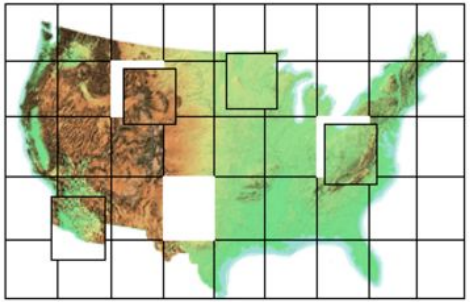


Storage Clusters (petabytes of data)
Computing clusters (1,000's of CPUs)



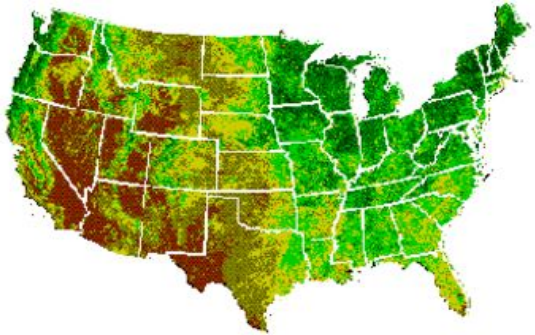
COMPUTATION EARTH ENGINE

Divides data in independent grids



Storage Clusters (petabytes of data)
Computing clusters (1,000's of CPUs)





Merges all grid calculations

COMPUTATION EARTH ENGINE

- Unprecedented speed: reduce processing times by orders of magnitude by using the distributed, cloud-based computing power
- Ease of use and lower costs: online platform with easy access to data, scientific algorithms, computational power

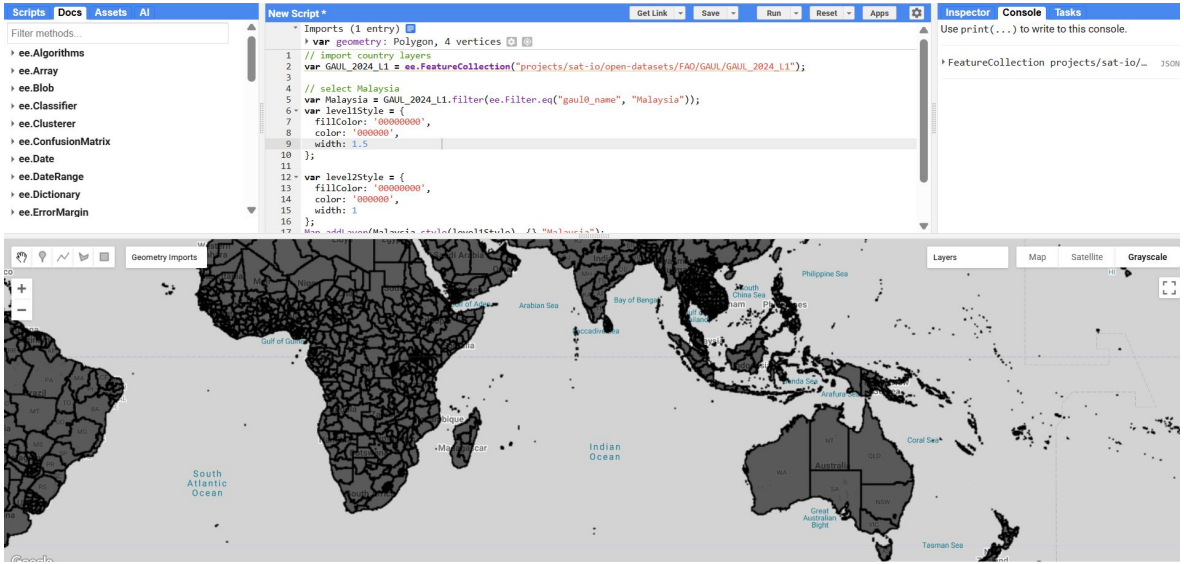
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CODE EDITOR

- What is it?
  - Web based IDE for the Earth Engine API
  - Access many pre-made geospatial tools



WHAT IS EARTH ENGINE?

What can Earth Engine do?

Get an image

Pick your: projection, resolution, bands, bounding-box, visualization



WHAT IS EARTH ENGINE?

What can Earth Engine do?

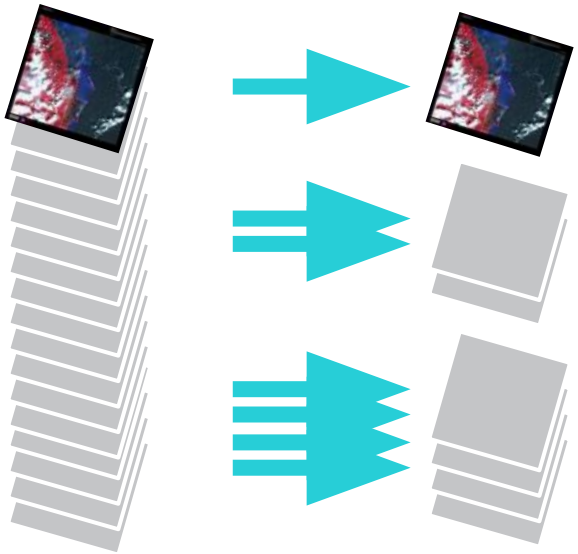
- Get an image
- Apply an algorithm to an image
- *Use library functions or script your own*



WHAT IS EARTH ENGINE?

What can Earth Engine do?

- Get an image
- Apply an algorithm to an image
- Filter a collection
- *Time, Space & Metadata Search*

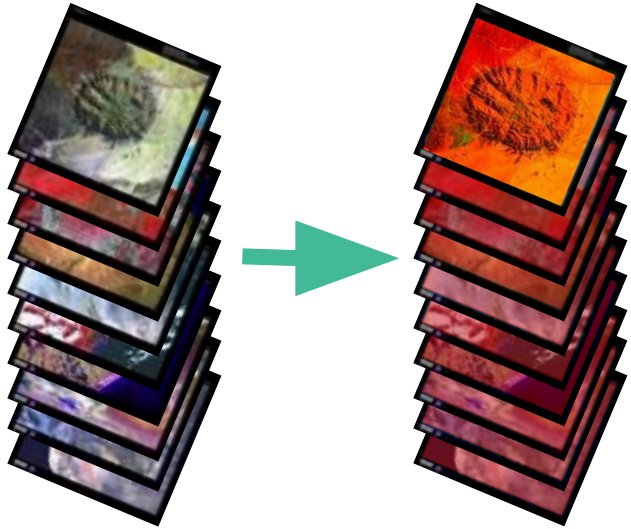


WHAT IS EARTH ENGINE?

What can Earth Engine do?

- Get an image
- Apply an algorithm to an image
- Filter a collection
- Map an algorithm over a collection

$N \rightarrow N$

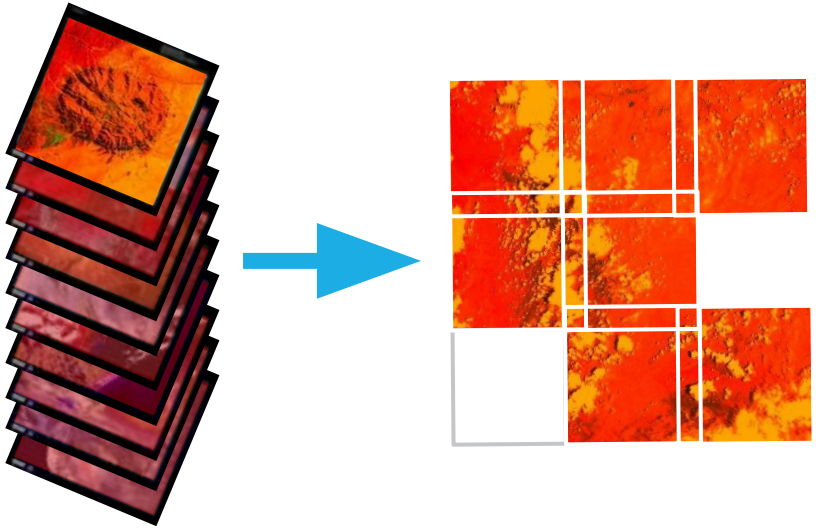


WHAT IS EARTH ENGINE?

What can Earth Engine do?

- Get an image
- Apply an algorithm to an image
- Filter a collection
- Map an algorithm over a collection
- Reduce a collection

$N \rightarrow 1$ or $N \rightarrow M$

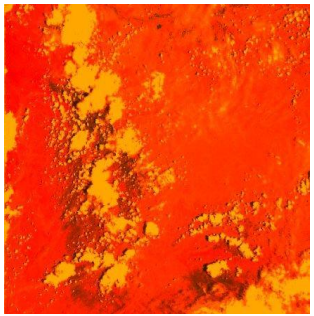


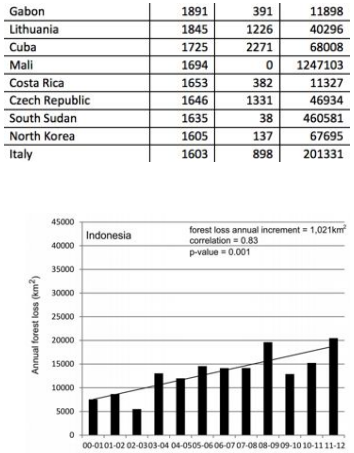
WHAT IS EARTH ENGINE?

WHAT IS EARTH ENGINE?

What can Earth Engine do?

- Get an image
- Apply an algorithm to an image
- Filter a collection
- Map an algorithm over a collection
- Reduce a collection



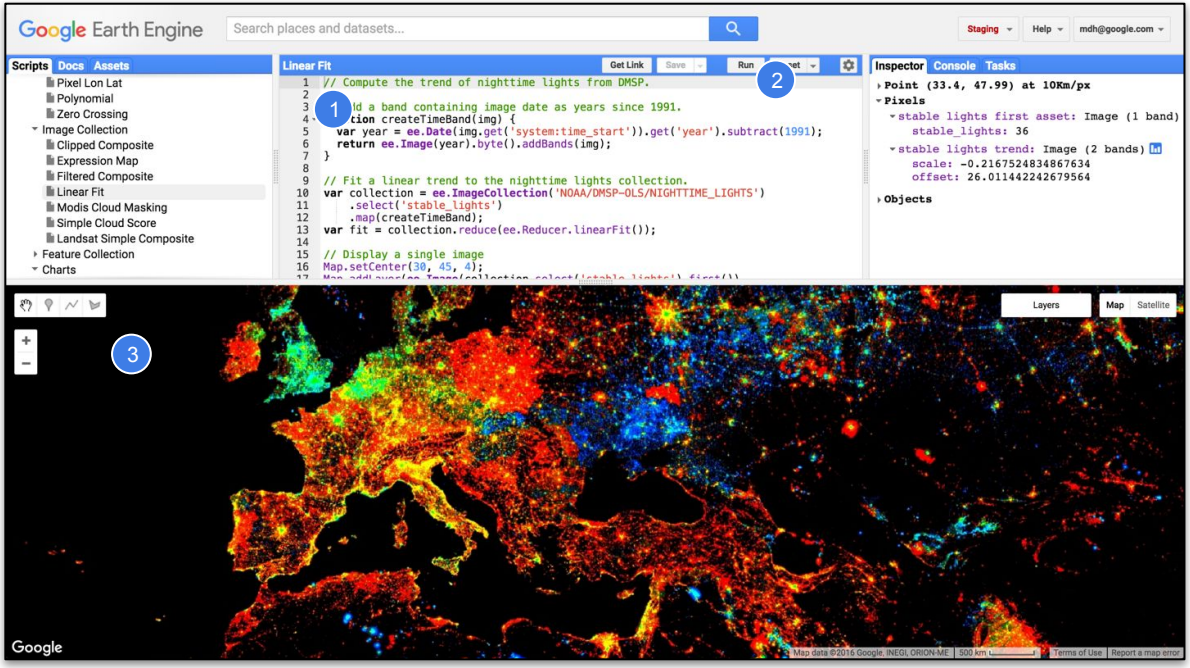


Gabon	1891	391	11898
Lithuania	1845	1226	40296
Cuba	1725	2271	68008
Mali	1694	0	1247103
Costa Rica	1653	382	11327
Czech Republic	1646	1331	46934
South Sudan	1635	38	460581
North Korea	1605	137	67695
Italy	1603	898	201331

CODE EDITOR INTERFACE

In interactive mode, the user can:

- Edit their script 1
- Run the script 2
- Evaluate the results 3
- Repeat, refine, and perfect
- Share and publish



code.earthengine.google.com

WHAT IS EARTH ENGINE?

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1. Public data catalog: vast amounts of publicly available data (you don’t need to store data)
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WHEN TO USE EARTH ENGINE

EE Benefits:

Good for projects that requires:

- Data coverage for a large region
- Extensive data library
- High speed, intensive processing capacity
- Advanced raster processing tools

EE Limitations:

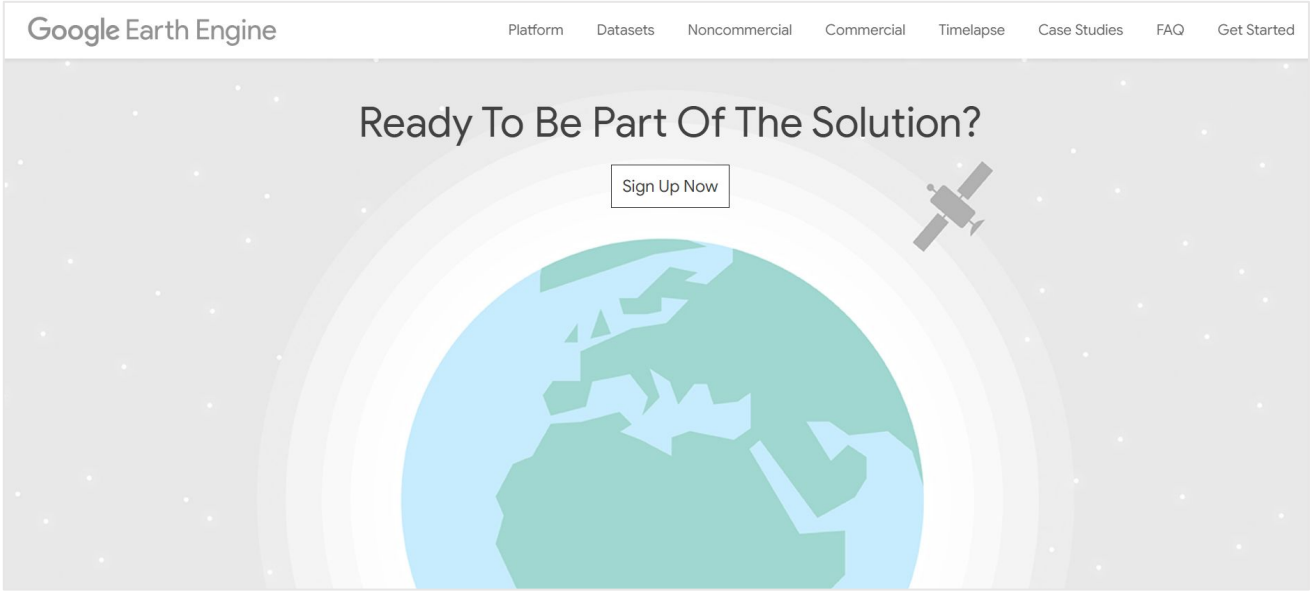
- Better suited to image analyses than vector-based analyses
- Analysis based on pixel spatial relations are harder to complete (because of the processing on multiple CPUs). Image segmentation and hydrologic modeling options are limited or in testing phases.

REGISTRATION PROCESS

Sign Up for GEE

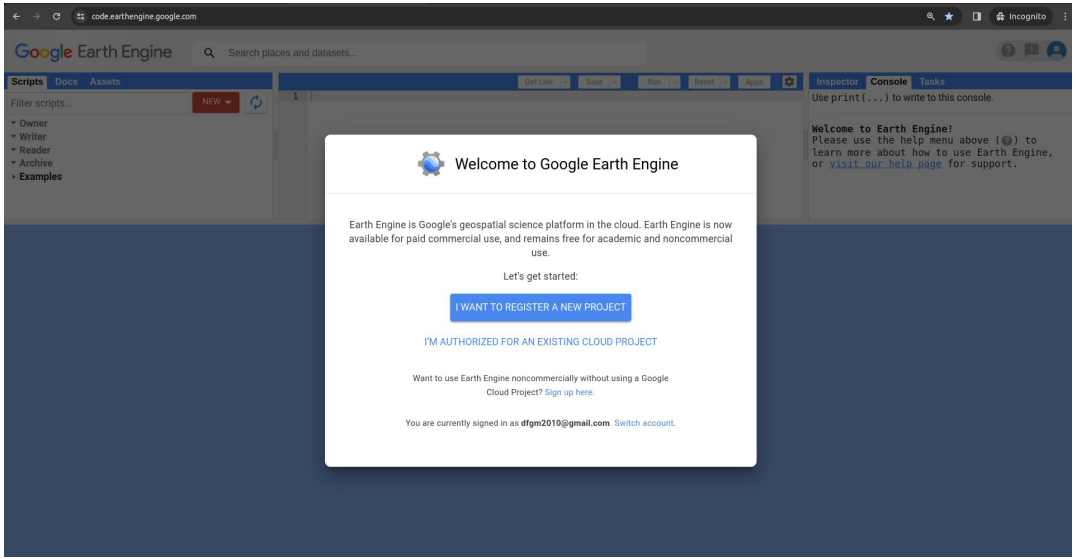


https://earthengine.google.com/signup/



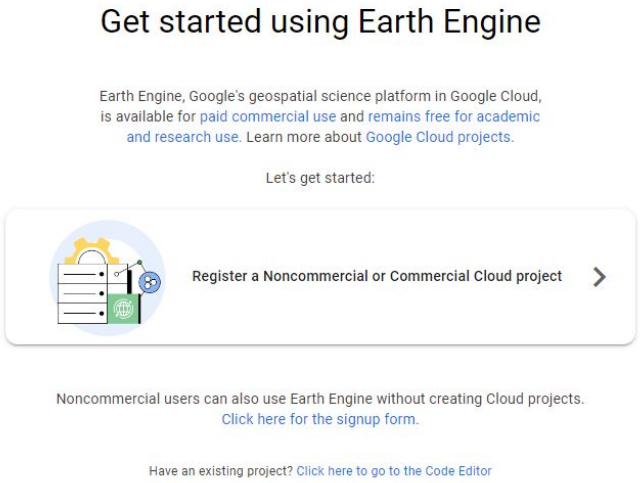
Sign up for GEE

- If you already have a Google email address, sign up for a GEE account by going to https://earthengine.google.com/new_signup.
- If you don't have a Google email address, sign up by going to http://mail.google.com/mail/sign-up.
- Once you have a GEE account, access the interface by going to https://code.earthengine.google.com.



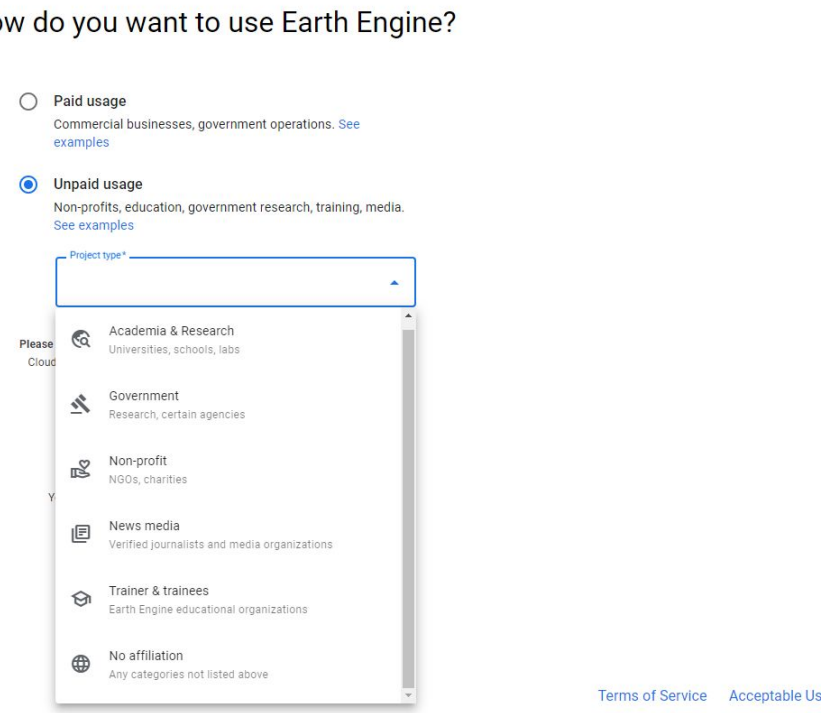
Sign up for GEE

- If you already have a Google email address, sign up for a GEE account by going to https://code.earthengine.google.com/register
- The image to the right is the first page you'll see
- Click on the 'Register a Noncommercial or Commercial Cloud project'



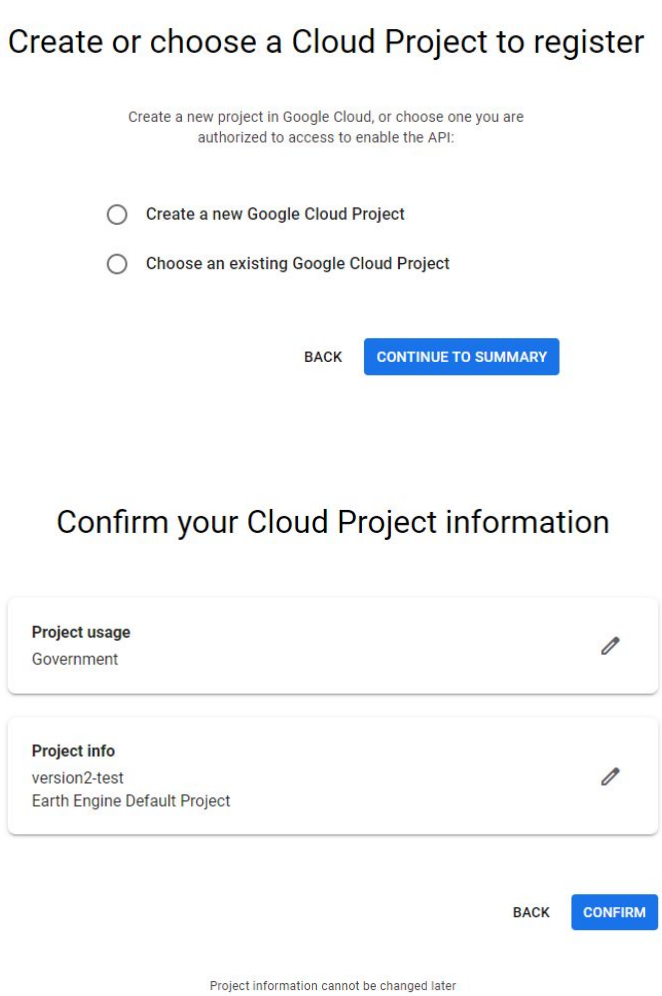
Sign up for GEE

- Select the **‘Trainer & trainees’** project type



Sign up for GEE

- Once you move to the next page,
- You can then create a new Google Cloud Project or join an existing one
- Click ‘Continue to Summary’
- If all the information looks correct, select ‘Confirm’



PART 2: CONTENT OUTLINE



INTRODUCTION TO THE CODE EDITOR API

web-based IDE for the Earth Engine JavaScript API



WORKING WITH IMAGES

Images are composed of one or more bands and each band has its own name, data type, scale, mask and projection



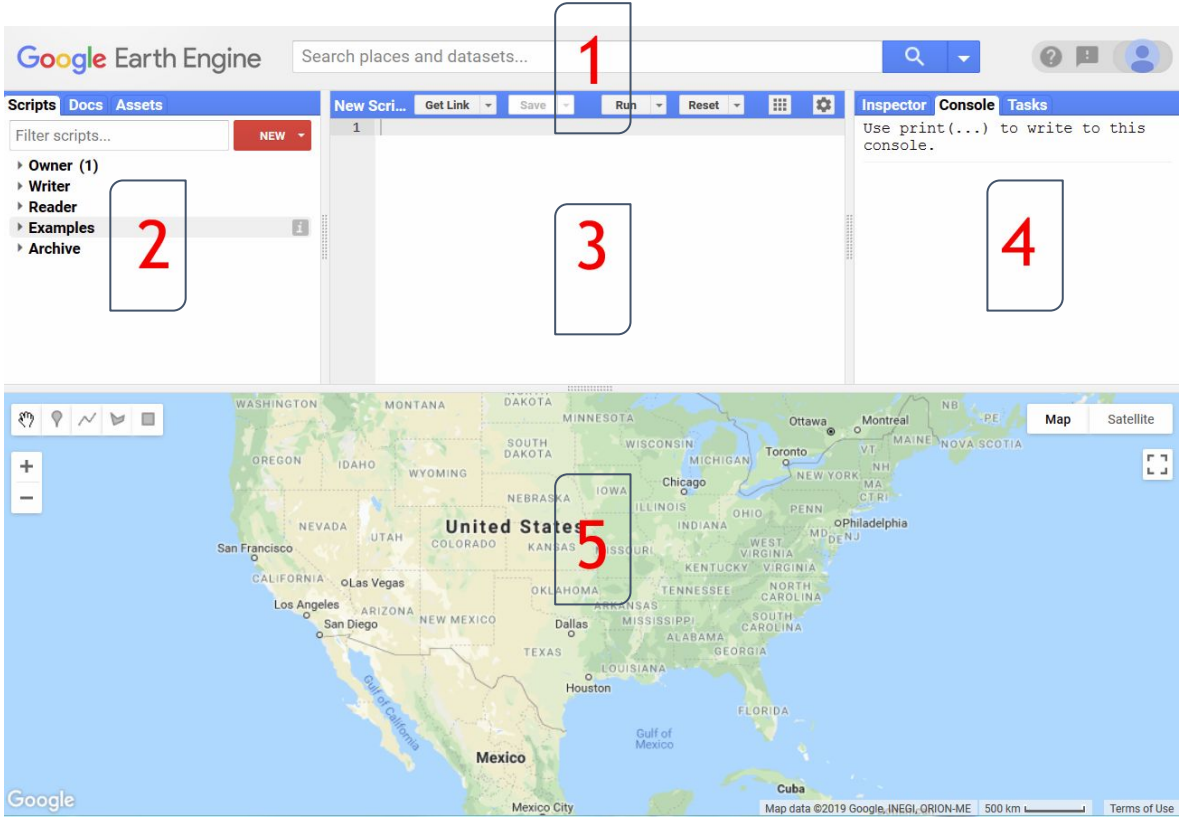
GEE Applications

Publicly available datasets

Examples of using GEE

NAVIGATING THE USER INTERFACE

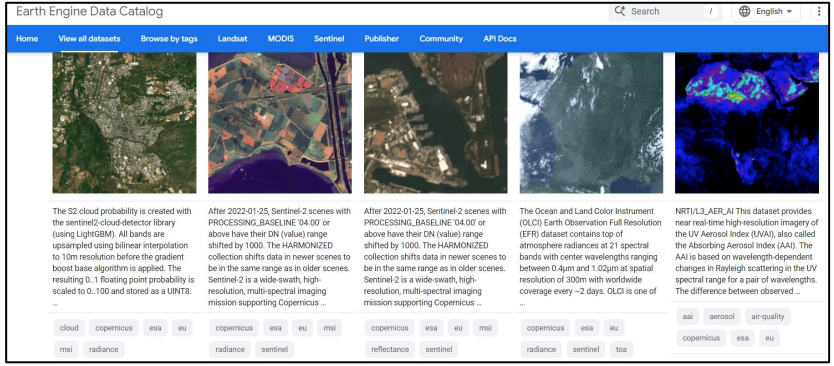
1. SEARCH BAR
2. LEFT PANEL
3. CODE EDITOR
4. RIGHT PANEL
5. MAP VIEWER

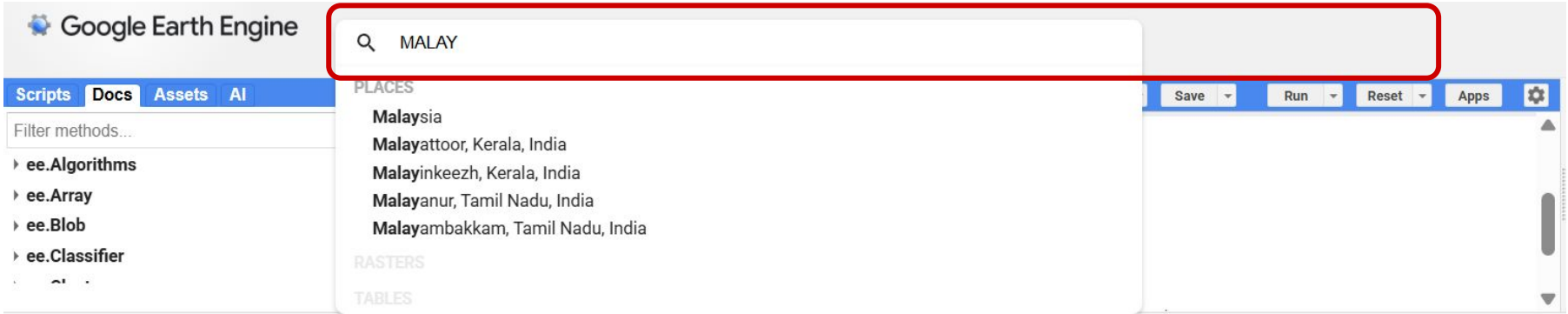


NAVIGATING THE USER INTERFACE

SEARCH BAR

This tool helps GEE users quickly access datasets and zoom to locations





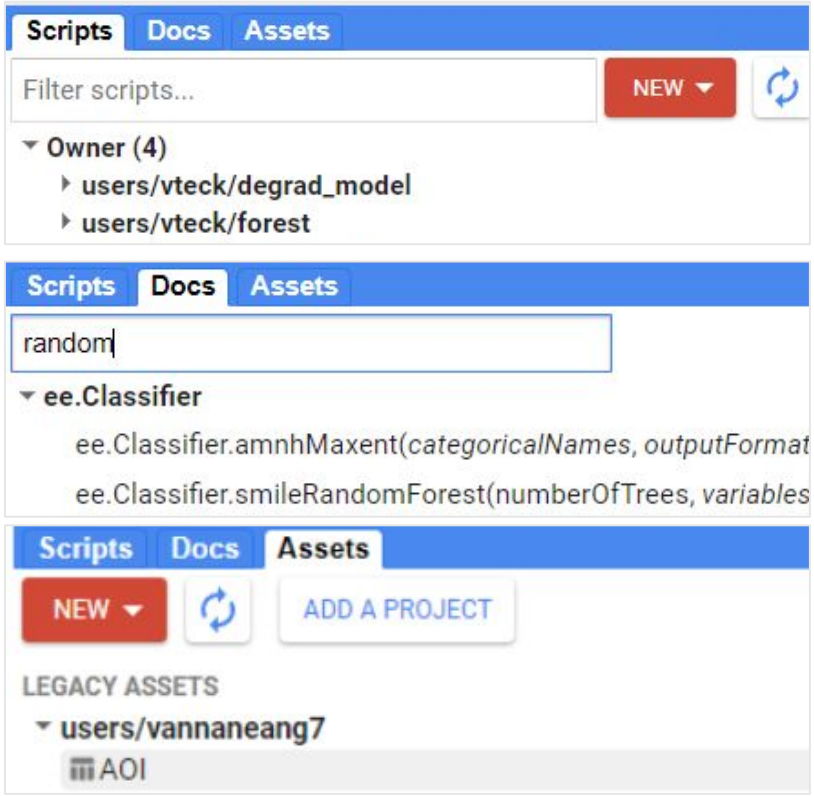
NAVIGATING THE USER INTERFACE

LEFT PANEL

SCRIPT: In this tab, any script you save will be listed under “Owner”.

DOCS: In this tab, you will see a categorical listing of available functions to use as you write your script.

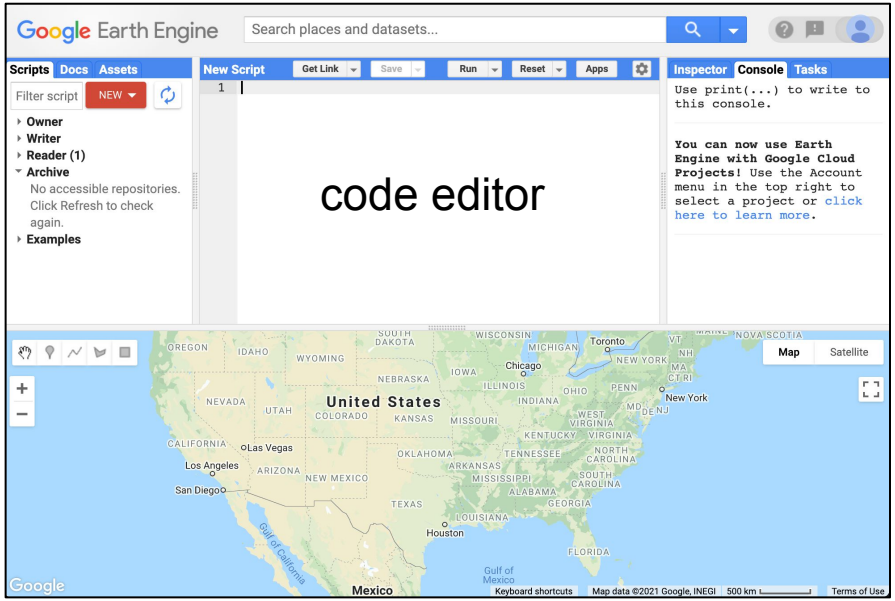
ASSETS: Any files that you have uploaded to or exported to GEE will be listed in this tab.



NAVIGATING THE USER INTERFACE

CODE EDITOR

All GEE scripts are written in javascript and can be activated by clicking the "**Run**" button or pressing **ctrl-enter** (cmd-enter on **Mac**). A word of caution: it is not automatically save!



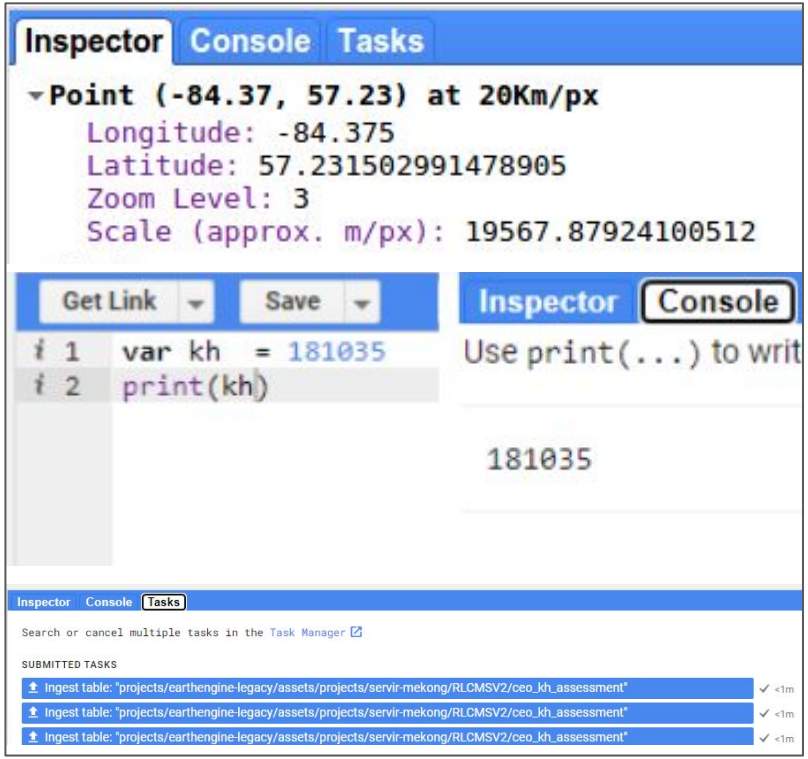
NAVIGATING THE USER INTERFACE

RIGHT PANEL

INSPECTOR: The mouse cursor changes to a crosshairs and clicking on the map allows us to sample our data from a particular **latitude** and **longitude.**

CONSOLE: This is the default tab when GEE loads and contains helpful information related to data being accessed in the script editor as well as **statistical printouts** and **error messages.**

TASKS: In this tab, we can track the progress of any data that are in the queue for **export/download.**



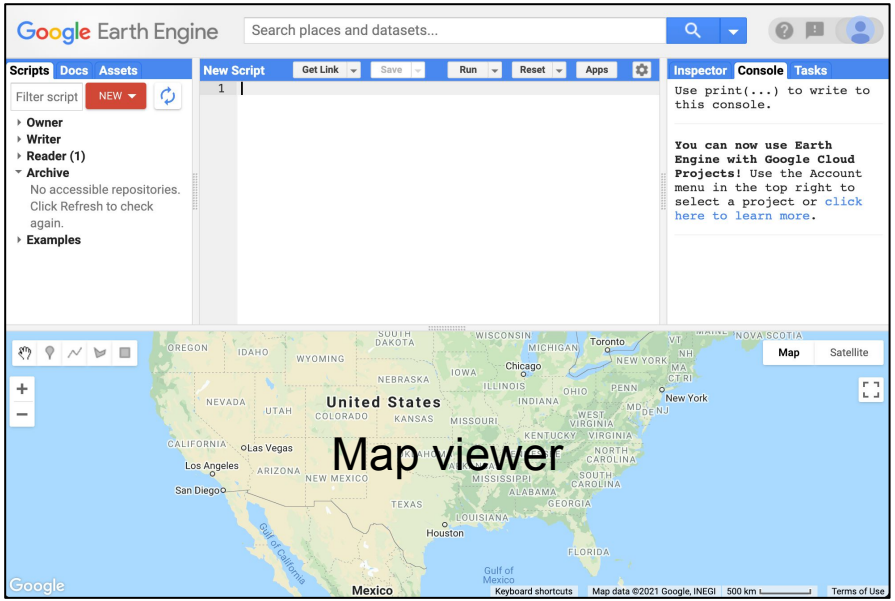
NAVIGATING THE USER INTERFACE

MAP VIEWER

GEOMETRY TOOL: (upper left corner of the map) The hand icon is the cursor for panning around the map. Users can then import geometries to your script by drawing them on the screen. You can draw geometries for:

- points
- lines
- polygons
- rectangles

LAYER PANEL: (upper right corner of the map) This drop down is used to adjust the display of layers you added to the map.



INTRO TO JavaScript in GEE

COMMENT

DOUBLE SLASH (//): for indicating a line of code is a comment, to explain the code or leave a note for yourself.

The Code Editor also provides a shortcut— Ctrl + / on Windows, Cmd + / on Mac —to comment or uncomment multiple lines at a time.

// This is comment!

```
print('Hello World');

// var city = 'San Francisco';
// print(city)

// var population = 873965;
// print(population);

// var cities = ['San Francisco', 'Los Angeles', 'New York', 'Atlanta'];
// print(cities);

// var cityData = {
//   'city': 'San Francisco',
//   'population': 873965,
//   'coordinates': [-122.4194, 37.7749]
// };

// print(cityData);

// var greet = function(name) {
//     return 'Hello ' + name;
// };
// print(greet('World'));
// print(greet('Readers'))

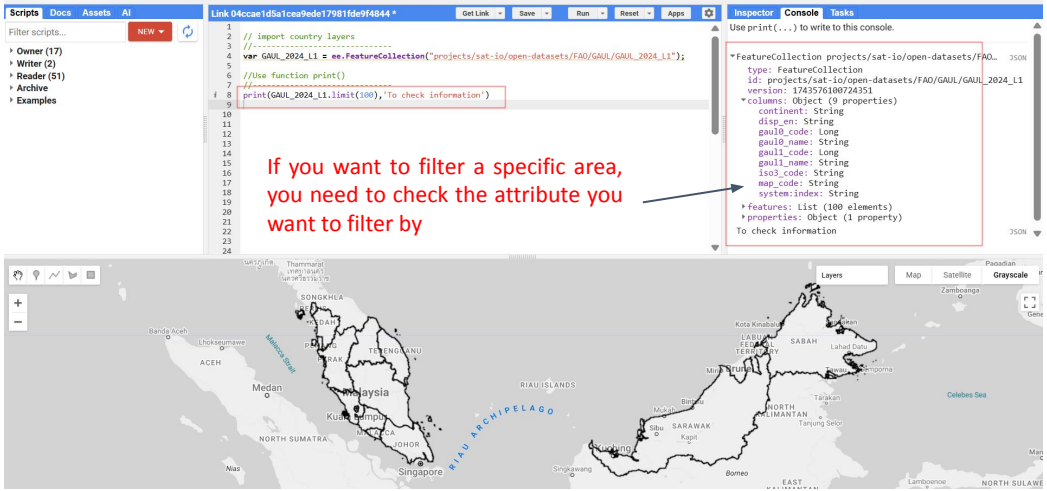
// This is a comment
```

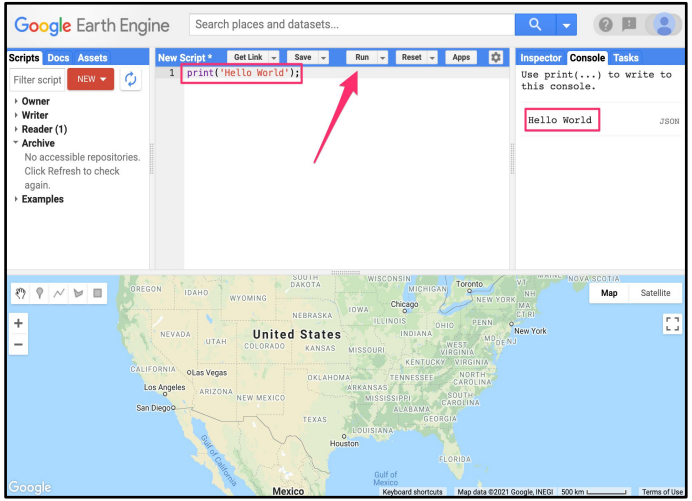
INTRO TO JavaScript in GEE

PRINT FUNCTION

Print is a function to print text or further information, where the output will be displayed on the upper right-hand panel under the Console after clicking the Run button

print('Hello World')





Let's go to: https://code.earthengine.google.com/ Let's try to display the words "**Hello World**"

INTRO TO JavaScript in GEE

VARIABLE (var)

A **variable** is a storage location in programming that holds data that can be changed or manipulated.

var city = 'Johor'

NOTE: The code below assigns the text "**Johor**" to the variable named **city. Single quotes (')** or **Double quotes (")** on either side of the text can be used to create a text string in the code.

LISTS

JavaScript provides a data structure called a **list** that can hold multiple values. We can create a new list using the square brackets **[ ]** and adding multiple values separated by a comma (,). This example is a list of strings of city names assigned to the variable cities.

```
var cities = ['Johor', 'Kedah', 'Kelantan', 'Melaka'];
print(cities);
```

Let's go to: https://code.earthengine.google.com/ Let's try to display "var" and list "[ ]"

CONTENT OUTLINES

OBJECTS

Objects in JavaScript allow you to store **key-value pairs**, where each value can be referred to by its key. You can create a dictionary using curly braces **{ }.** The code below creates an object called cityData with some information about Johor.

```
var cityData = {
    'city': 'Johor',
    'coordinates': [103.7593, 1.4773],
    'population': 35,919,306
};
print(cityData);
```

FUNCTION

Functions take user inputs, use them to carry out some computation, and send an output back. Functions allow you to group a set of operations together and repeat the same operations with different parameters without having to rewrite them every time.

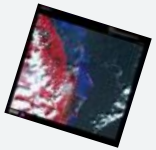
```
var greet = function(name) {
    return 'Hello ' + name;
};
print(greet('World'));
print(greet('Readers'));
```

ASSET TYPES

RASTER

Image

- A single raster image
- May have multiple bands, properties*



ImageCollection

- Indexed during creation
- Supports filtering and aggregation of many images by time, area, etc.



TABLE

FeatureCollection

- Geometry column inferred
  - Empty MultiPoint if missing
- Other columns → Feature properties
- Property/column keys: table.getInfo().columns



* All collections & elements can have key→value properties

ASSET TYPES

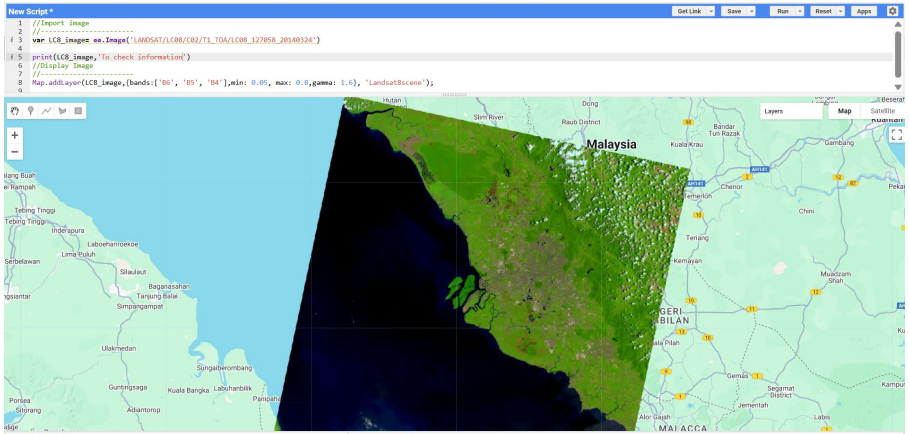
LANDSAT

Image (ee.Image) for Landsat

Fundamental raster data type for choosing an individual image in Earth Engine.

```
var LC8_image= ee.Image(
'LANDSAT/LC08/C02/T1_TOA/LC08_127058_20140324')

//Display Image
Map.addLayer(LC8_image,{bands:['B6', 'B5', 'B4'],min: 0.05, max: 0.8,gamma: 1.6}, 'Landsat8scene');
```



Code	Description
LANDSAT	Satellite mission family
LC08	Landsat 8 (Operational Land Imager/TIRS)
C02	Collection 2 (latest standard collection)
T1_TOA	Tier 1, Top of Atmosphere reflectance
LC08_127058_20140324	Scene ID
127	WRS-2 Path
58	WRS-2 Row
20140324	Acquisition date: March 24, 2014

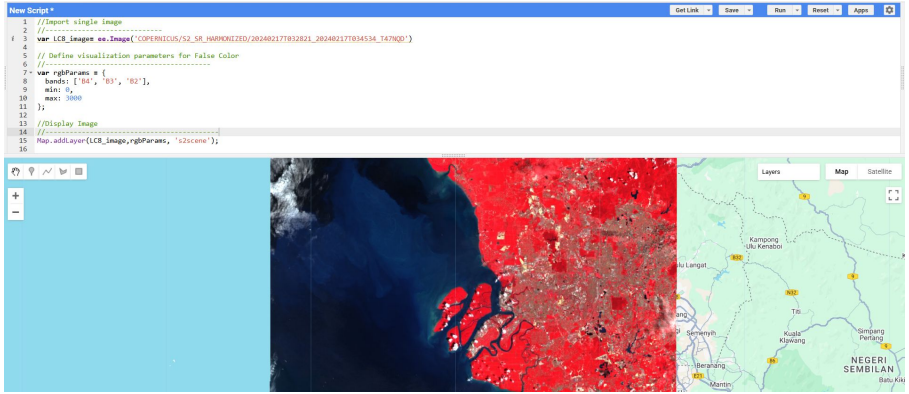
ASSET TYPES

SENTINEL-2

Image (ee.Image) for Sentinel-2

Fundamental raster data type for choosing an individual image in Earth Engine.

```
//Import single image
//----------------------------
var LC8_image=
ee.Image('COPERNICUS/S2_SR_HARMONIZED/20240217T032821_20240217T034534_T47NQD')
// Define visualization parameters for False Color
//----------------------------------------
var rgbParams = {
  bands: ['B4', 'B3', 'B2'],
  min: 0,
  max: 3000
};
//Display Image
//-----------------------------------------
Map.addLayer(LC8_image,rgbParams, 's2scene');
```



Component	Description
COPERNICUS/S2_SR_HARMONIZED	Sentinel-2 Harmonized Surface Reflectance collection
20240217T032821	Start time of image acquisition (2024-02-17, 03:28:21 UTC)
20240217T034534	End time of image acquisition (2024-02-17, 03:45:34 UTC)
T47NQD	Sentinel-2 Tile ID (UTM zone 47, latitude band N, grid square QD)

ASSET TYPES

LANDSAT

ImageCollection (ee.ImageCollection)

A stack of satellite images you can filter and process for cloud-free composites or time-series analysis

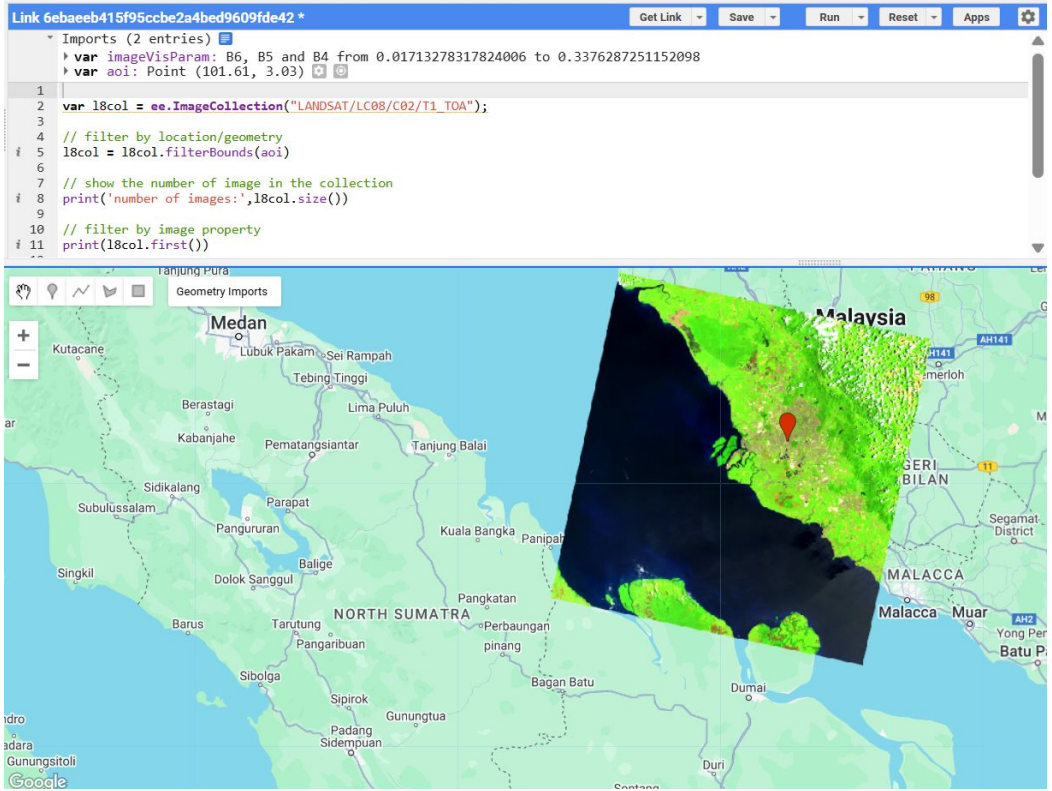
```
var l8col =
ee.ImageCollection("LANDSAT/LC08/C02/T1_TOA")
;

// filter by location/geometry
l8col = l8col.filterBounds(aoi)

// show the number of image in the collection
print('number of images:',l8col.size())

// filter by image property
print(l8col.first())

// filter by the least cloud cover
var cloudfreeImg = l8col.sort("CLOUD_COVER",
true).first() // true, false
print('cloud free image',cloudfreeImg)
```



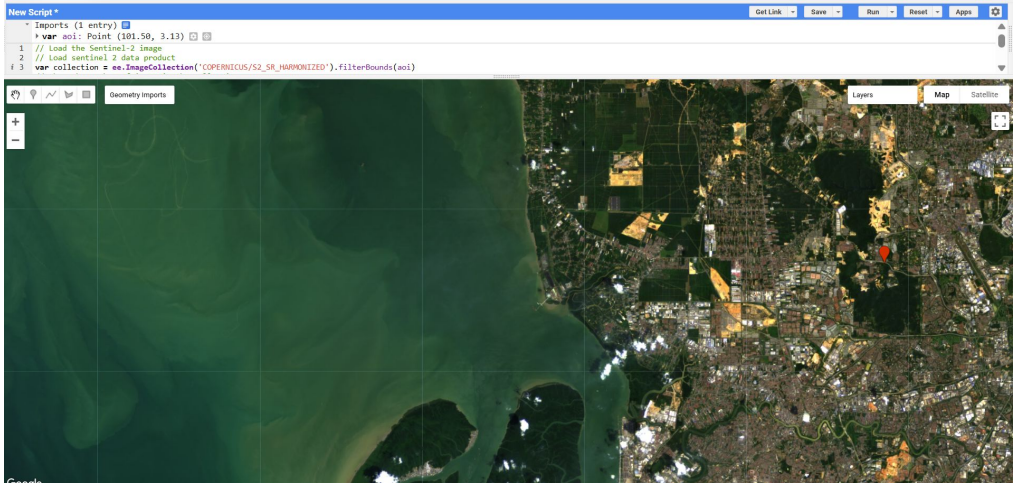
ASSET TYPES

SENTINEL-2

ImageCollection (ee.ImageCollection)

A stack of satellite images you can filter and process for cloud-free composites or time-series analysis

```
// Load sentinel 2 data product
var collection =
ee.ImageCollection('COPERNICUS/S2_SR_HARMONIZED').filterBounds
(aoi)
// show the number of image in the collection
print('number of images:',collection.size())
// filter by image property
print(collection.first())
var startingDate = '2024-01-01';
var endDate = '2024-12-31'
// filter by dates
var s2_2024 = collection.filterDate(startingDate,endDate)
print('2024 images:', s2_2024)
// filter by the least cloud cover
var cloudfreeImg = s2_2024.sort("CLOUDY_PIXEL_PERCENTAGE",
true).first() // true, false
print('cloud free image',cloudfreeImg)
// Define visualization parameters for RGB
var rgbParams = {
  bands: ['B4', 'B3', 'B2'],
  min: 0,
  max: 3000
};
//Display image
Map.addLayer(cloudfreeImg,rgbParams,'cloud free image')
```



📍 Location (`filterBounds`)

📅 Date (`filterDate`)

🌥️ Cloud cover or other metadata

ASSET TYPES

FeatureCollection (ee.FeatureCollection)

A set of Features.

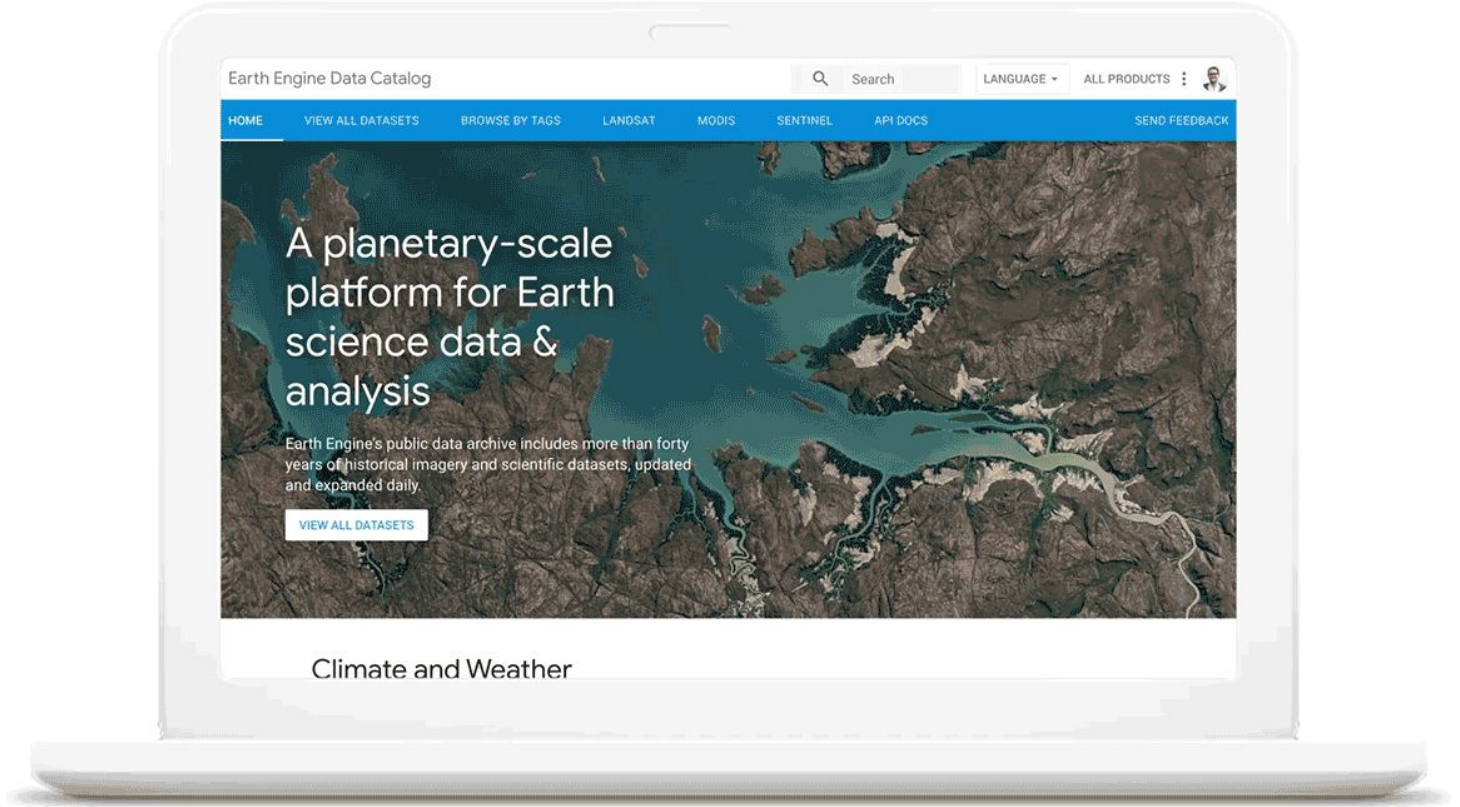
```
// import country layers
//------------------------------
var GAUL_2024_L1 =
ee.FeatureCollection("projects/sat-io/open-datasets/FAO/GAUL/GAUL_2024_L1");
//Use function print()
//------------------------------
print(GAUL_2024_L1.limit(100),'To check information')

// select Malaysia
//------------------------------
var Malaysia = GAUL_2024_L1.filter(ee.Filter.eq("gaul0_name", "Malaysia"));
print(Malaysia)
var level1Style = {
  fillColor: '00000000',
  color: '000000',
  width: 1.5
};
var level2Style = {
  fillColor: '00000000',
  color: '000000',
  width: 1
};
Map.addLayer(Malaysia.style(level1Style), {},"Malaysia");
```



Concept	Earth Engine Equivalent	Explanation
Rows	**Features**	Each feature is like one row — a single geographic object (e.g., a polygon for Malaysia).
Columns	**Properties / Attributes**	Each feature has properties (like "**gaul0_name**", "**gaul1_name**", "**area**") which are columns holding data.

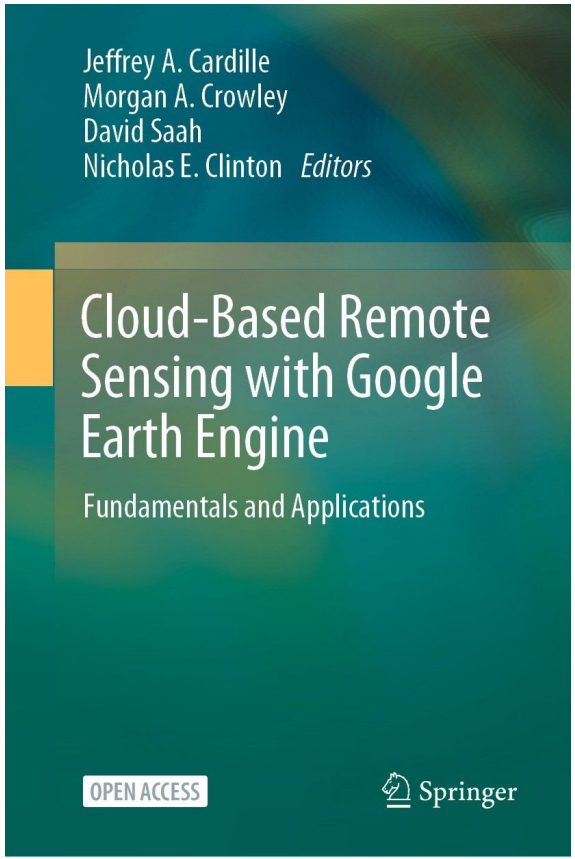
APPLICATIONS OF GEE



Go to developers.google.com/earth-engine/datasets

Resources

efabook



CO-EDITED BY SIG MANAGING PRINCIPAL DAVID SAAH
PUBLISHED ONLINE IN FALL OF 2022
100 CONTRIBUTORS - 8 SIG STAFF

The book is broadly organized into two sections:

1) **Fundamentals**: a set of 31 labs designed to take the reader from an Earth engine novice to advanced user;

2) **Applications**: tour of the world of Earth Engine showing how it is used in a variety of real-word settings that rely on remote-sensing data.

Thank You

Questions & Discussion

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